Deepak Kumar Ahirwar

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	5.33
jawahar navodaya vidyalaya varanasi UP	Class XII (CBSE)	2023	87.40%
jawahar navodaya vidyalaya panna MP	Class X (CBSE)	2021	87.80%

SCHOLASTIC ACHIEVEMENTS

NTSE Scholar (National Talent Search Examination) -- Awarded by the Government of India for exceptional academic performance; selected through a highly competitive two-stage national examination testing analytical ability, mathematics, and science aptitude

Dakshana Scholar (Dakshana Foundation) -- Selected for a fully funded, merit-based academic program focused on rigorous training in mathematics and science, aimed at nurturing high-potential students from diverse backgrounds

National Children's Science Congress (NCSC) -- Represented at the Regional Level for a research project on **Sustainable Energy**, involving study and analysis of renewable energy solutions, their efficiency, and real-world applicability

SKILLS

Programming & Data Skills: Python, SQL, Data Analysis, Data Preprocessing, Data Cleaning, Data Handling, Feature Engineering, Feature Extraction, Time-Series Analysis, Machine Learning

Tools & Technologies: Microsoft Excel, Pandas, NumPy, Matplotlib, Jupyter Notebook, Git, Github, LaTeX

Analytical Skills: Anomaly Detection, Root Cause Analysis, Data Cleaning, Performance Monitoring

OS: Windows, Ubuntu , Arch.

Soft Skills: Technical Communication, Problem Solving, Cross-team Collaboration, Structured Thinking

RELEVANT COURSES

Core Electrical Engineering Courses: Electrical Machines, Power Systems Analysis, Power Electronics, Control Systems, Signal Processing, Communication Systems, Analog Electronics, Semiconductor Device Fundamentals, Engineering Electromagnetics, Circuits and Network Analysis

Mathematics Courses: Probability Theory, Random Processes, Differential Equations and Transform Techniques, Vector Calculus, Matrix Theory, Mathematics for Electrical Engineers

Programming & Data Courses: Scientific Programming, C Programming, Data Structures and Algorithms, Machine Learning

PROJECTS

AI-Based High-Precision Industrial Thermometer

- Designed a **PT100 RTD-based temperature sensing system** using **Arduino Uno** with **Wheatstone bridge + 16-bit ADS1115 ADC** for high-accuracy measurement.
- Applied **ML regression models (Quadratic LSQ, SGD, Random Forest)** for **calibration and non-linearity compensation**.
- Built an **OCR-based automated data pipeline (EasyOCR, OpenCV)** for synchronized dataset generation from a reference thermometer.
- Achieved **high accuracy (MSE 0.009, R^2 0.9999)** using **Quadratic LSQ** and deployed it on **embedded hardware** for real-time prediction.

Scientific Calculator using Arduino (Scientific Computing on Hardware)

- Implemented a scientific calculator on **ATmega328P (Arduino Uno)** using **numerical methods** for evaluating algebraic and transcendental functions
- Applied **Runge-Kutta (RK4)** and **difference equations** for accurate computation of **trigonometric, logarithmic, and exponential functions**
- **Developed an expression parser using the Shunting Yard algorithm** for efficient evaluation of mathematical expressions
- Integrated **keypad input** and **LCD display** for real-time computation on **resource-constrained hardware**

Electricity Consumption Analysis & Forecasting using Time-Series Data

- Performed **data preprocessing** on **time-series data**, including **timestamp parsing** and handling missing values
- Conducted **exploratory data analysis (EDA)** to identify **usage patterns**, **peak demand periods**, and relationships with environmental factors
- Engineered **time-based features** such as **rolling averages** and **daily trends** to improve model performance
- Built **regression models** for **short-term forecasting** and evaluated performance using standard metrics
- Visualized **consumption trends** and **model predictions** using Python **Matplotlib**

EXTRACURRICULAR

- Participated in **Youth Parliament**, developing **public speaking, debate, and critical thinking** skills through policy discussions
- Active volunteer at **National Service Scheme (NSS)**, contributing to **community development and social impact initiatives**